arm AGI CPU

Disaggregated, chiplet-based server SoC to enable scalable coherency and memory bandwidth in the terabyte/s era

Deepak Goel
Vice President of Solutions Engineering
Arm

Saurabh Pradhan
Director of Product Management
Arm

arm NEOVERSE

1.5B

Neoverse cores shipped into datacenters

500M in last 9 months

>60%

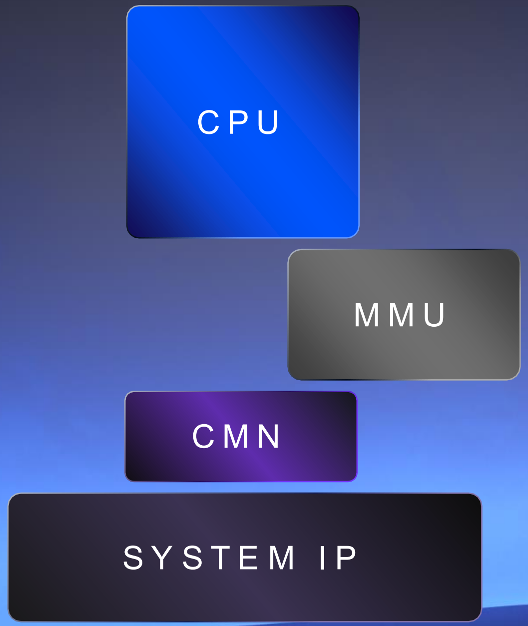
Arm share as host CPU in AI server platforms*

*IDC Report 2026

>120k

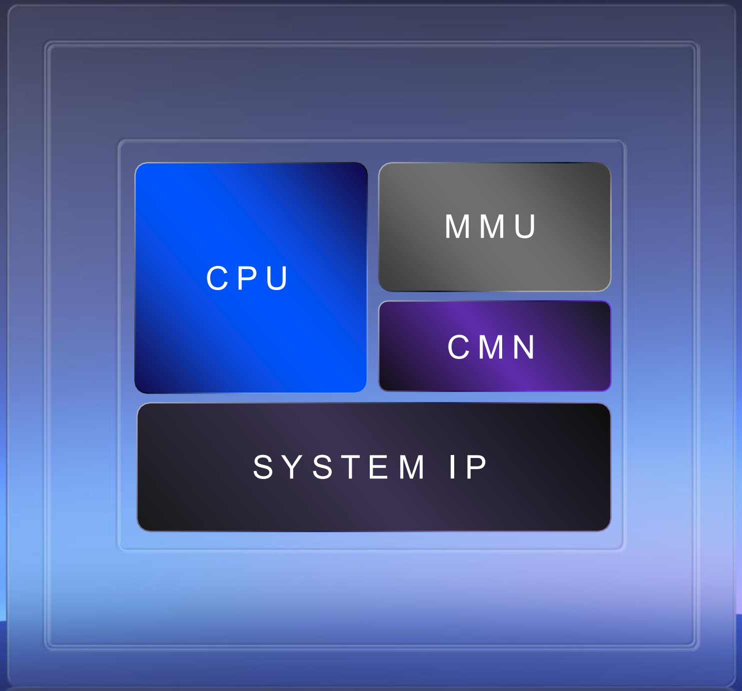
Companies deploying Arm in the cloud today

Arm Neoverse



IP

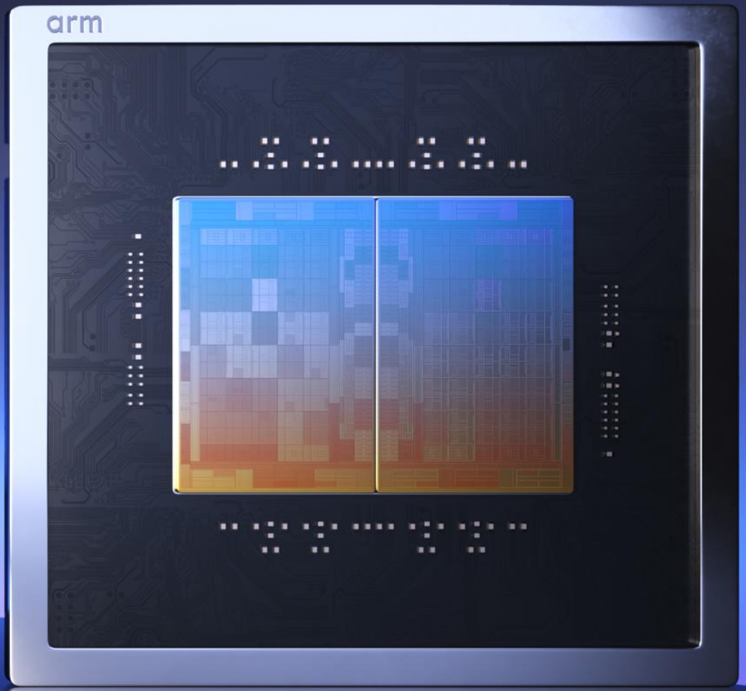
+



Compute Subsystems (CSS)

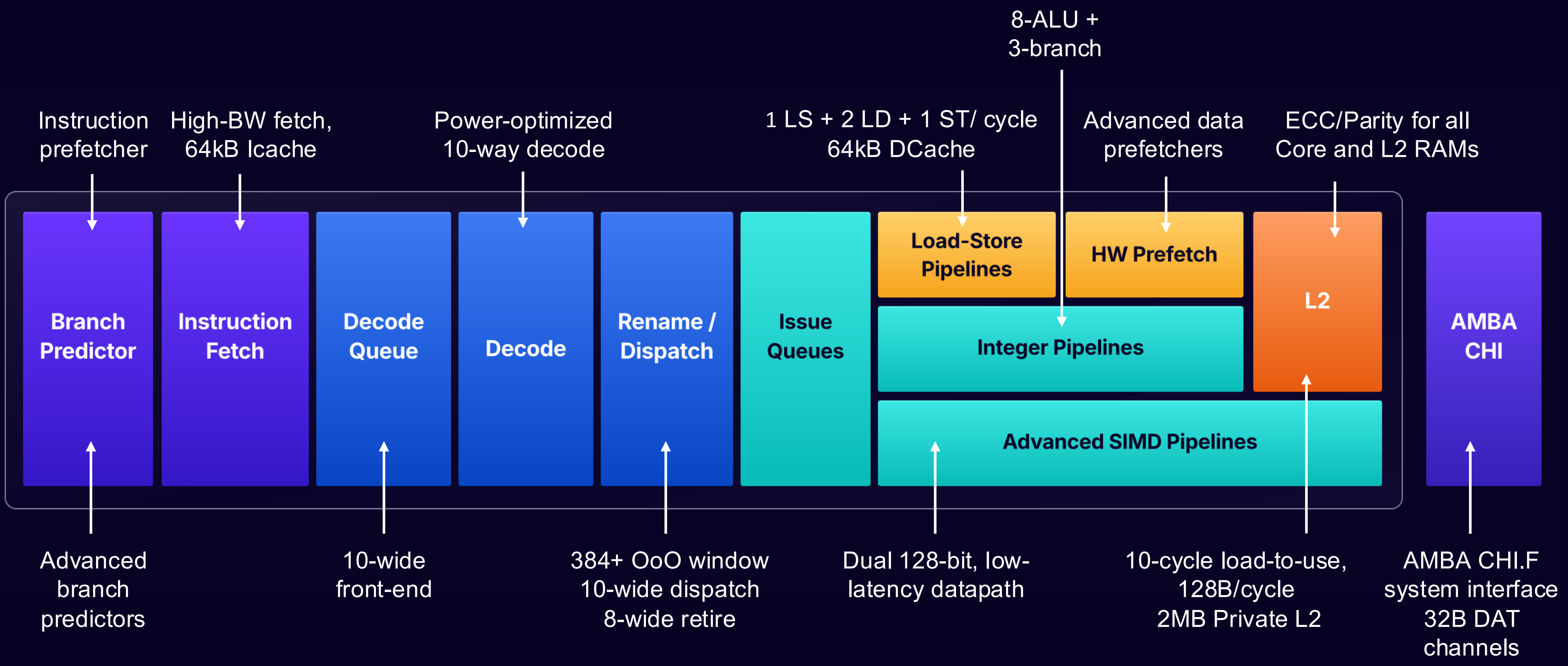
+

arm AGI CPU



First production silicon

Arm Neoverse V3 core



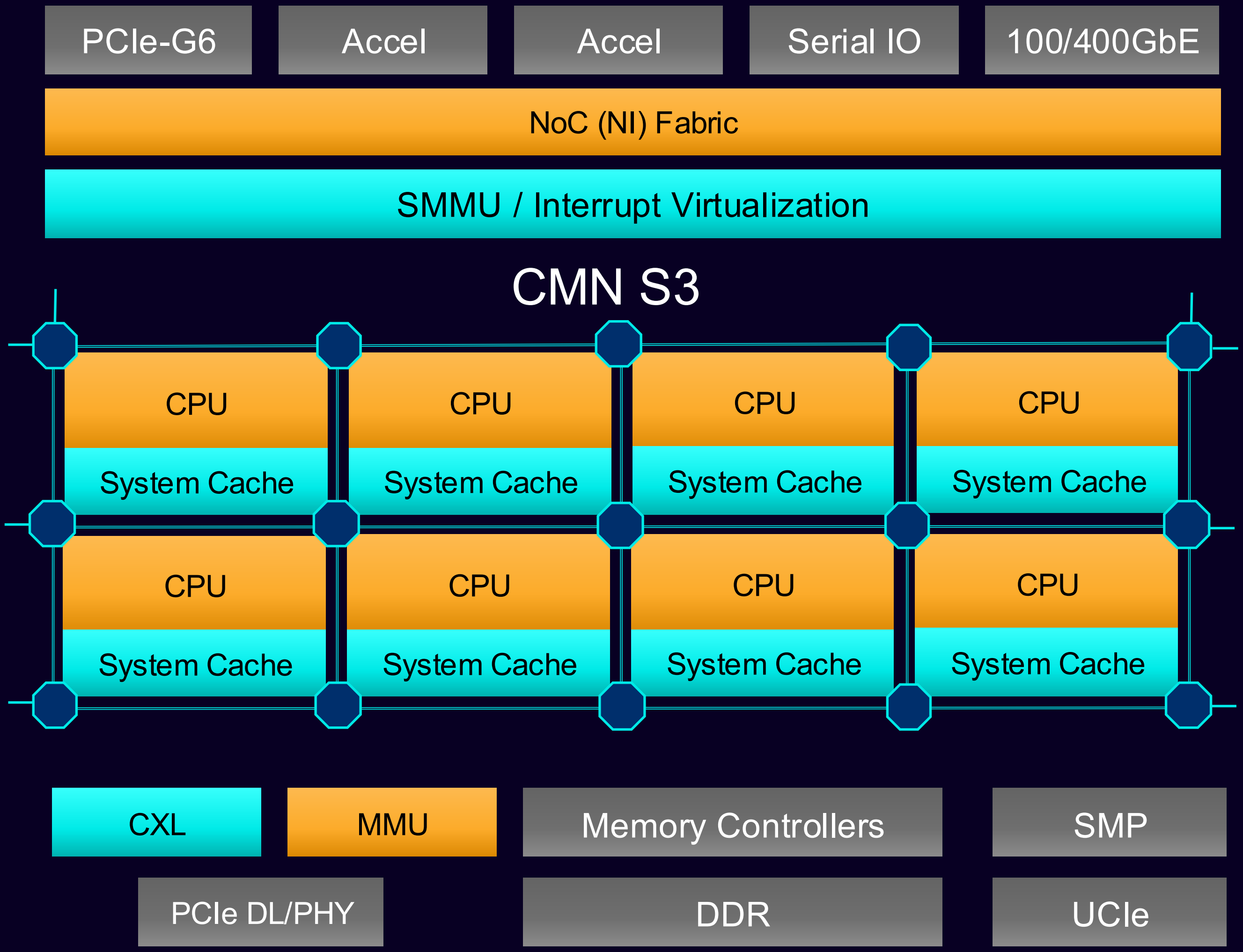
CMN-S3: Arm Coherent Mesh Network

Scalable coherent fabric and hierarchical caching

- 8 rows x 9 columns CMN-S3 mesh delivers high-bandwidth, low-latency coherency across CPUs, memory, I/O, and accelerators.
- 128 MB distributed SLC with snoop filtering and HN-S hierarchical caching supports advanced CHI-F coherency and RAS.

System scale-up, memory expansion, and high-speed IO

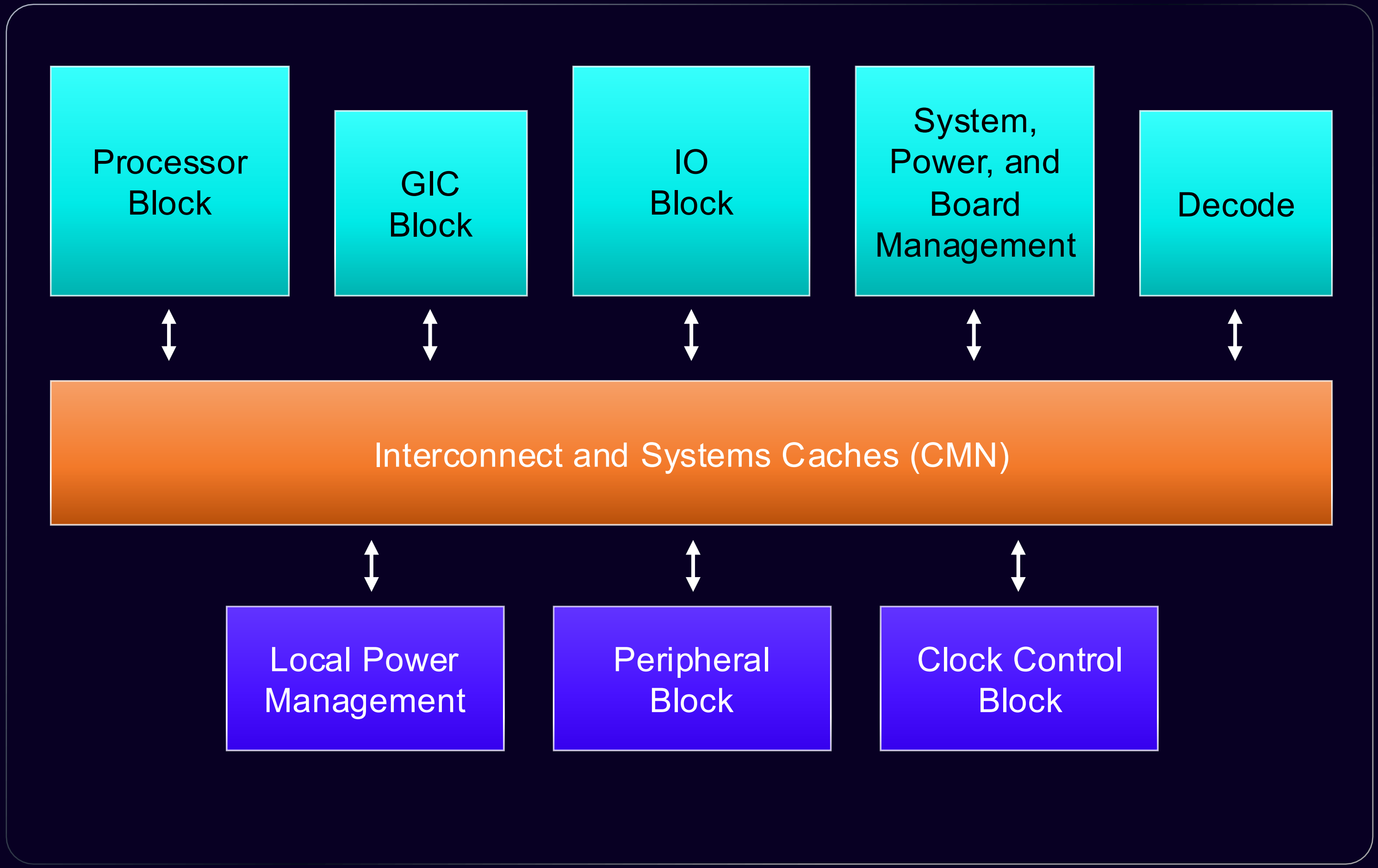
- Coherent multi-chiplet scaling, two-socket SMP, and CXL 3.0 Type 3 memory expansion.
- Per socket: 96 PCIe Gen6 lanes and 12 DDR controllers for high-bandwidth I/O and memory access.



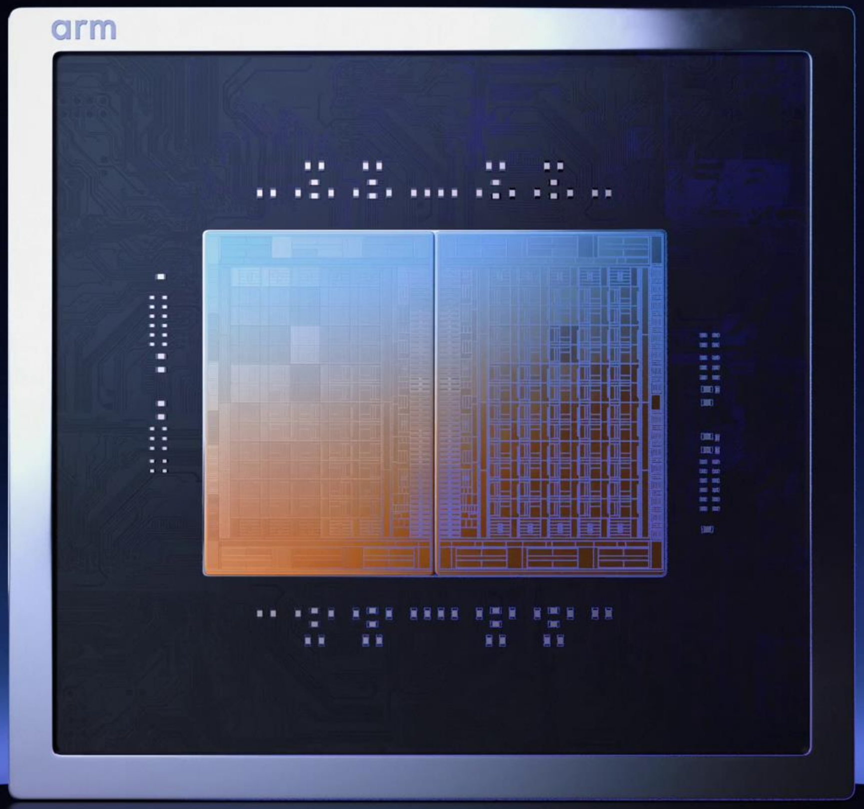
Arm CSS V3

The foundation of Arm AGI CPU

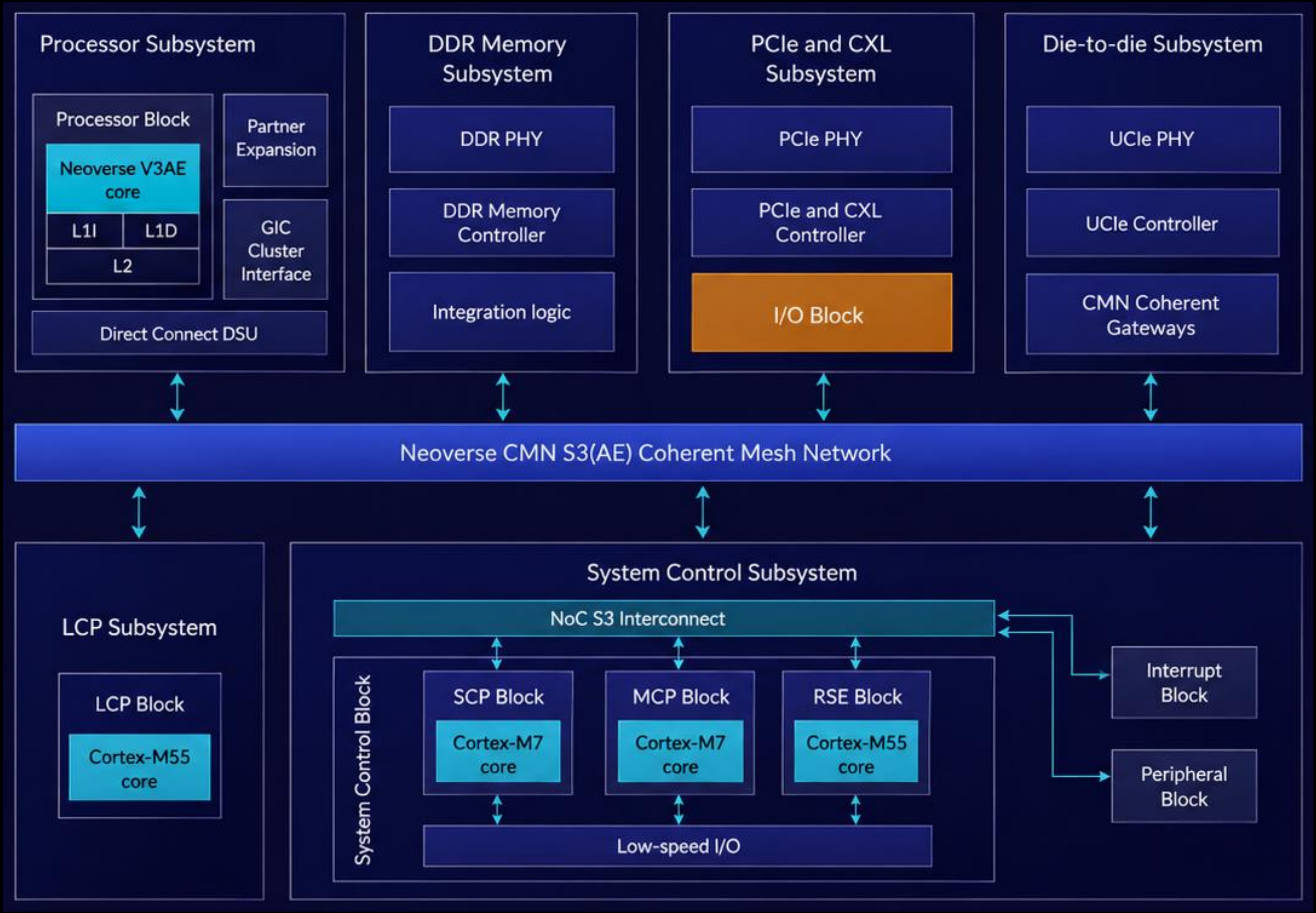
- 70 Arm Neoverse V3 cores per chiplet
  - 64KB I-cache/D-cache/Core
  - 2MB L2 private cache/Core
- Coherent CMN interconnect
  - 128MB shared system level cache
- Dual Socket support (2 chiplets per socket)
- Runtime security Engine as RoT
- System Control Processor (SCP) for system level power management
- Local Control Processors (LCP) for local power management
- Manageability Control processor (MCP) as satellite manageability core to interface with external BMC
- IO Virtualization support with SMMU-700



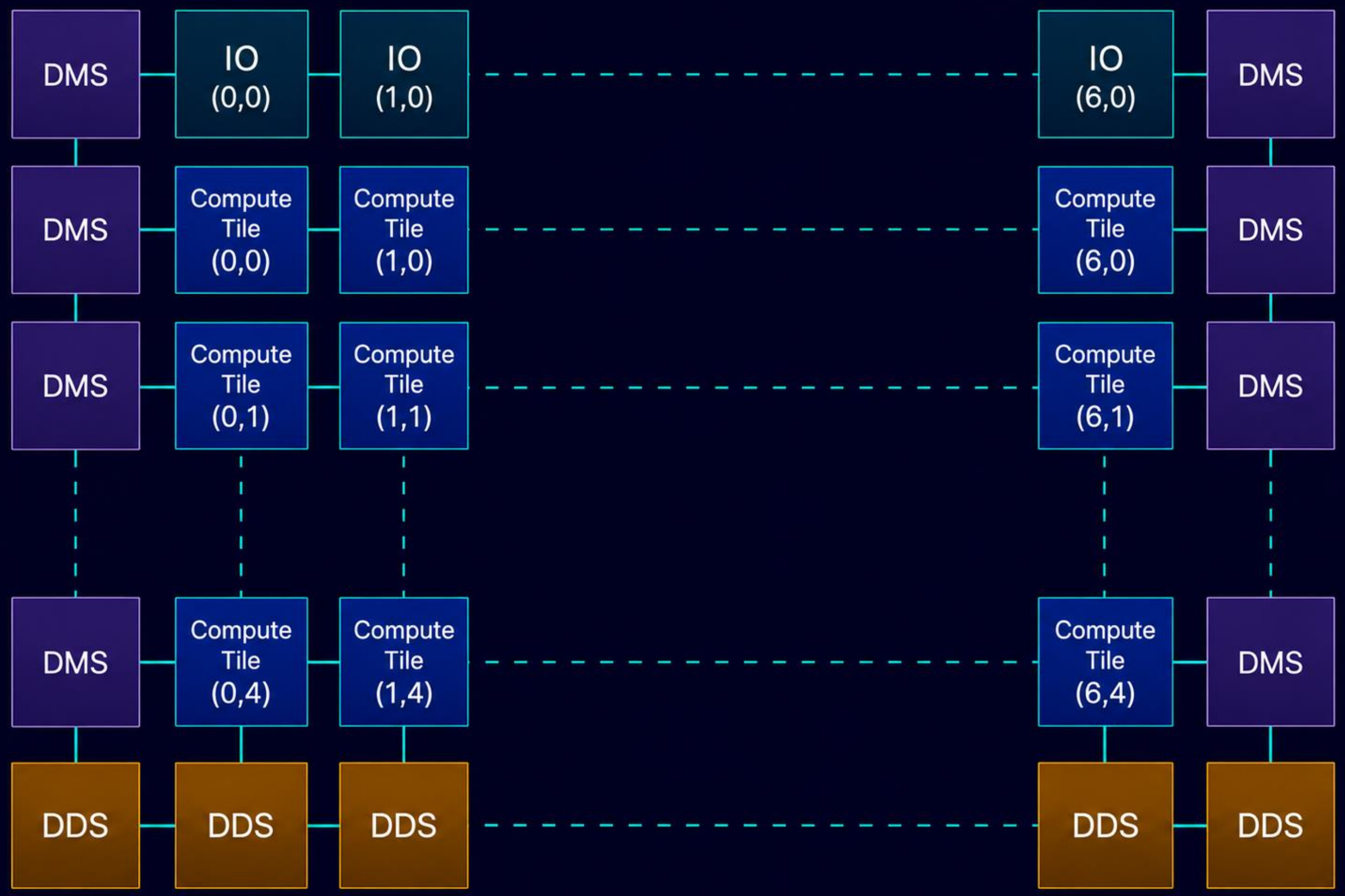
arm AGI CPU



Single chiplet block diagram



AGI CPU chiplet interconnect diagram



AGI CPU floorplan

Technology:

TSMC N3P

Legend:

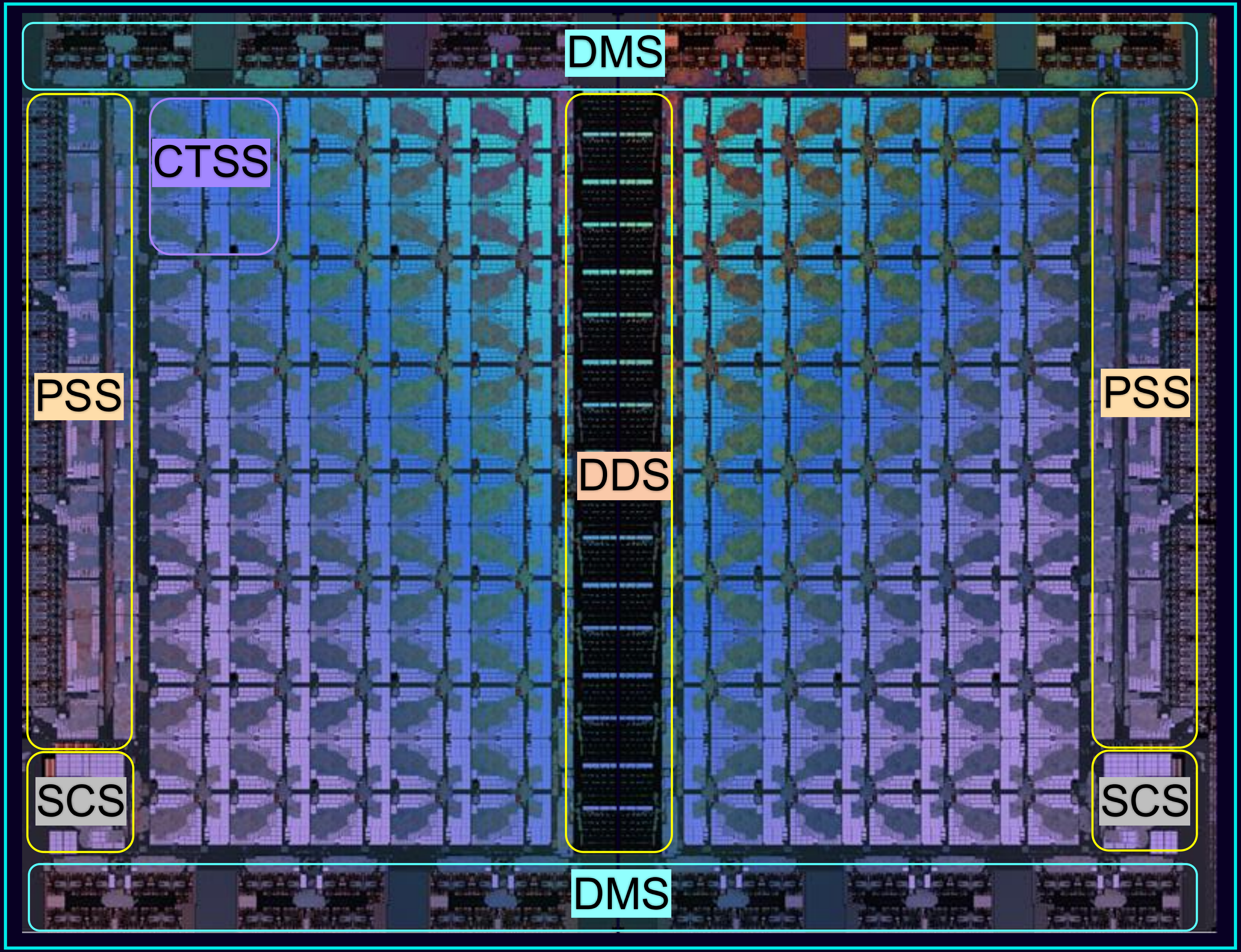
CTSS : Compute subsystem

DDS : Die-to-Die subsystem

DMS : DDR memory subsystem

SCS : System control subsystem

PSS : PCIe subsystem



SoC

arm

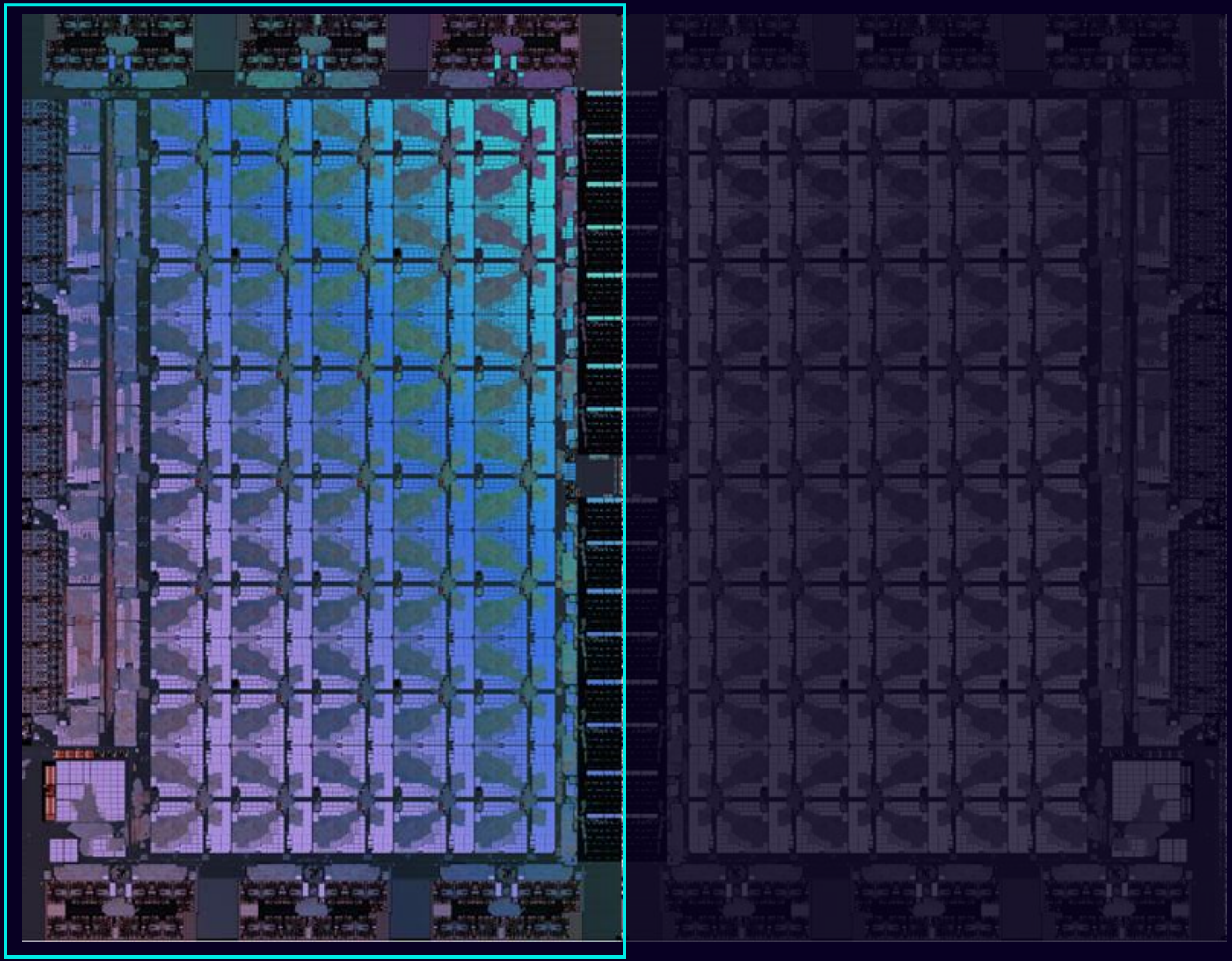
AGI CPU floorplan

Technology:

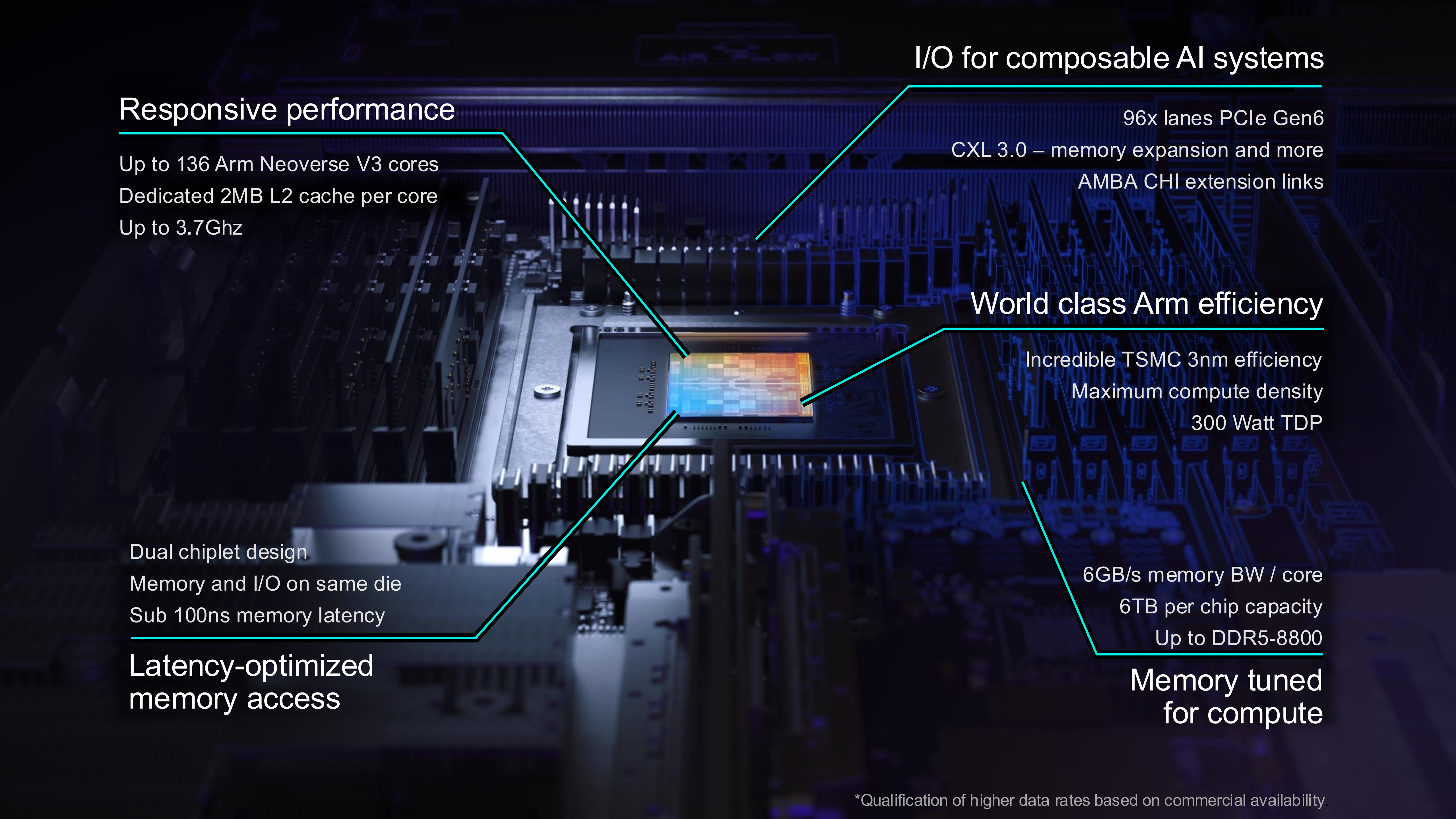
TSMC N3P

Transistors per chiplet:

>50B



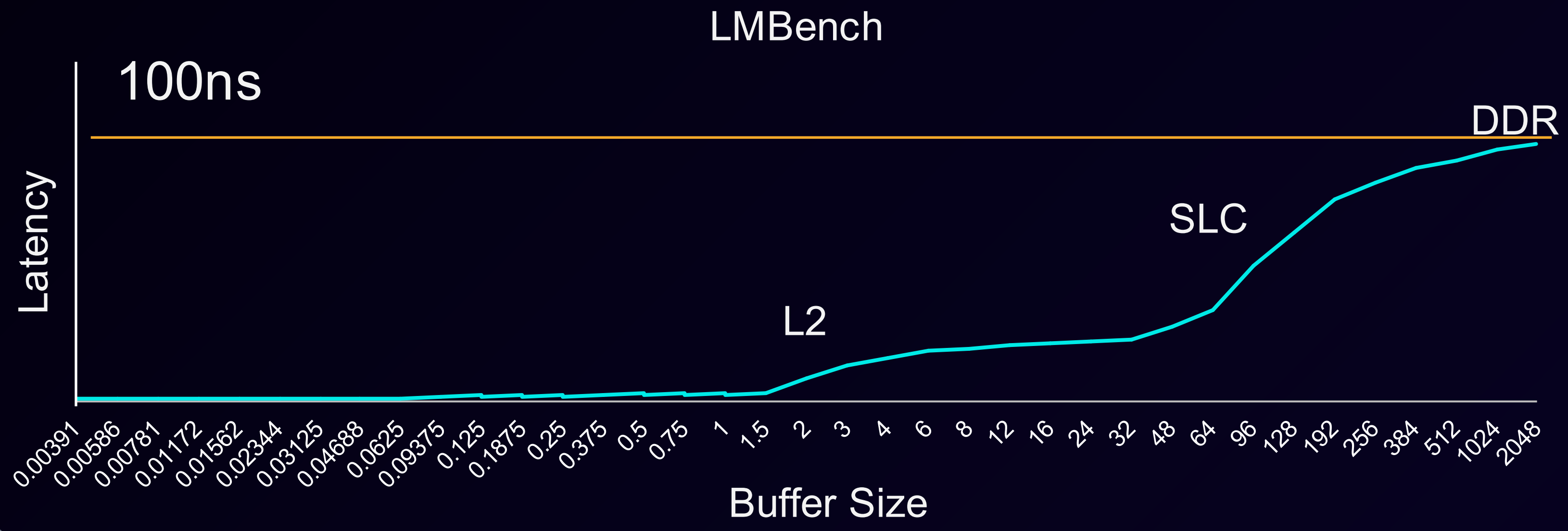
Chiplet

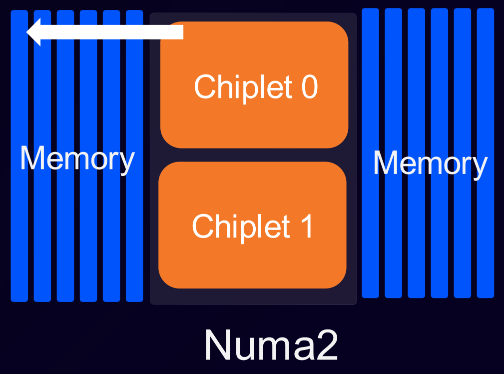


Responsive performance
Up to 136 Arm Neoverse V3 cores
Dedicated 2MB L2 cache per core
Up to 3.7Ghz
I/O for composable AI systems
96x lanes PCIe Gen6
CXL 3.0 – memory expansion and more
AMBA CHI extension links
World class Arm efficiency
Incredible TSMC 3nm efficiency
Maximum compute density
300 Watt TDP
Dual chiplet design
Memory and I/O on same die
Sub 100ns memory latency
Latency-optimized memory access
6GB/s memory BW / core
6TB per chip capacity
Up to DDR5-8800
Memory tuned for compute
*Qualification of higher data rates based on commercial availability

Elements of efficient performance

IPC	Tracking to production targets	300W
Frequency	Tracking to production targets	
Memory BW	DDR5-8000 validated for production*	
Memory Latency	<100ns	





arm

*Qualification of higher data rates based on commercial availability.

Arm AGI CPU: Features and specs

Feature type	Instance count	Type/features	Aggregated capacity/throughput
Processing cores	64 cores	64 KB I Cache 64 KB D Cache 2 × 128-bit vector; Crypto and RME supported; MTE2, MTE3, and ETEv1p2 not supported	Typical frequency: 2.8 GHz Nominal frequency: 3.5 GHz Boost frequency: 3.7 GHz
	128 cores		Typical frequency: 2.8 GHz Nominal frequency: 3.2 GHz Boost frequency: 3.5 GHz
	136 cores		
Level 2 Cache (L2S)	64 cores	2 MB per core	128 MB
	128 cores		256 MB
	136 cores		272 MB
System-level Cache (SLC)	64 cores	Remote caching (super-home node)	128 MB
	128 cores		
	136 cores		
PCIe and CXL	96 lanes	Gen6-capable; bifurcation: 1 ×16, 2 ×8, 4 ×4, 1 ×8, or 2 ×4. CXL bifurcation: 2 ×8. CXL 3.0 Type 3 and CML_SMP support.	768 GBps (Tx) + 768 GBps (Rx) (simultaneously active)
PCIe low-speed lanes	6 lanes	×1 link (Gen4-capable)	12 GBps (Tx) + 12 GBps (Rx)
DDR5	12 × 80 channels	RDIMM • 1 DIMM per channel, 2 ranks at 8800 Mbps(1) • 2 DIMMs per channel, 4 ranks at 6400 Mbps	Aggregated raw throughput: • ~845 GBps at 8800 Mbps(1) • ~614 GBps at 6400 Mbps
UCIe standard(2)	16 macros (×16)	BER: 1e-15 or less; 32 Gbps data-lane transfer rate	1 TBps (Tx) + 1 TBps (Rx) (simultaneously active)
I3C	6 used + 4 reserved	Supports BMC interfacing, external DIMM telemetry reading, and external socket interfacing	Typical frequency: 1 MHz
I2C	3 used + 5 reserved	SMBus-capable I2C interface; supports BMC interfacing and PCIe hot-plug detection	Typical frequency: 400 kHz–1 MHz
SPI	2	Interface to boot flash (QSPI) and TPM	Typical frequency: 50 MHz

1. Qualification of higher data rates based on commercial availability.
2. UCIe-S links are used for chiplet-to-chiplet connection. The SoC does not provide external UCIe ports.

arm

Memory subsystem, RAS and power management

Memory subsystem

- DDR5 up to 8800 MT/s, Fully out-of-order command scheduling, anti-starvation mechanisms, address mapping for bank parallelism and programmable page policies
- Bandwidth control: Provides memory-bandwidth limiting and monitoring through MPAM, QoS-based traffic prioritization and congestion feedback to regulate traffic

Memory subsystem (RAS)

- DDC / Single-DRAM-device failure correction (Chipkill-class protection), memory scrubbing, row-hammer mitigation, repair support, error injection and RAS error records

Hierarchical power and thermal management

- Local Control Processor (Arm Cortex M55) for a column of cores in a chiplet
- System Control Processor (Arm Cortex M7) per chiplet
- Shared voltage rail/chiplet architecture

arm

PCIE subsystem & Die-to-die subsystem

PCIE subsystem

- Total PCIe lanes (96 Gen6 + 6 Gen4 Utility Lanes)
- Supports CXL Type3 devices with no back invalidate
- Six PCIe Subsystems (PSS)
- Lane bifurcation: x16, 2 x8, 4 x4 for x16 PSS
- PCIe Hot-Plug support
- RAS: AER, DPC, ECRC, poison handling, surprise link down recovery

Die-to-die subsystem

- Standard Packaging: UCIe standard compliant. Sixteen macros@32Gbps per chiplet
- Use CCG (CMN Coherent Gateway) to connect two Phoenix Compute Chiplets (PCC)
- Latency optimized interface for communication across two chiplets

Arm AGI CPU 1OU dual node reference server

Customer reference server

PROCESSOR
- 2x Arm AGI CPU
- Up to 272 cores

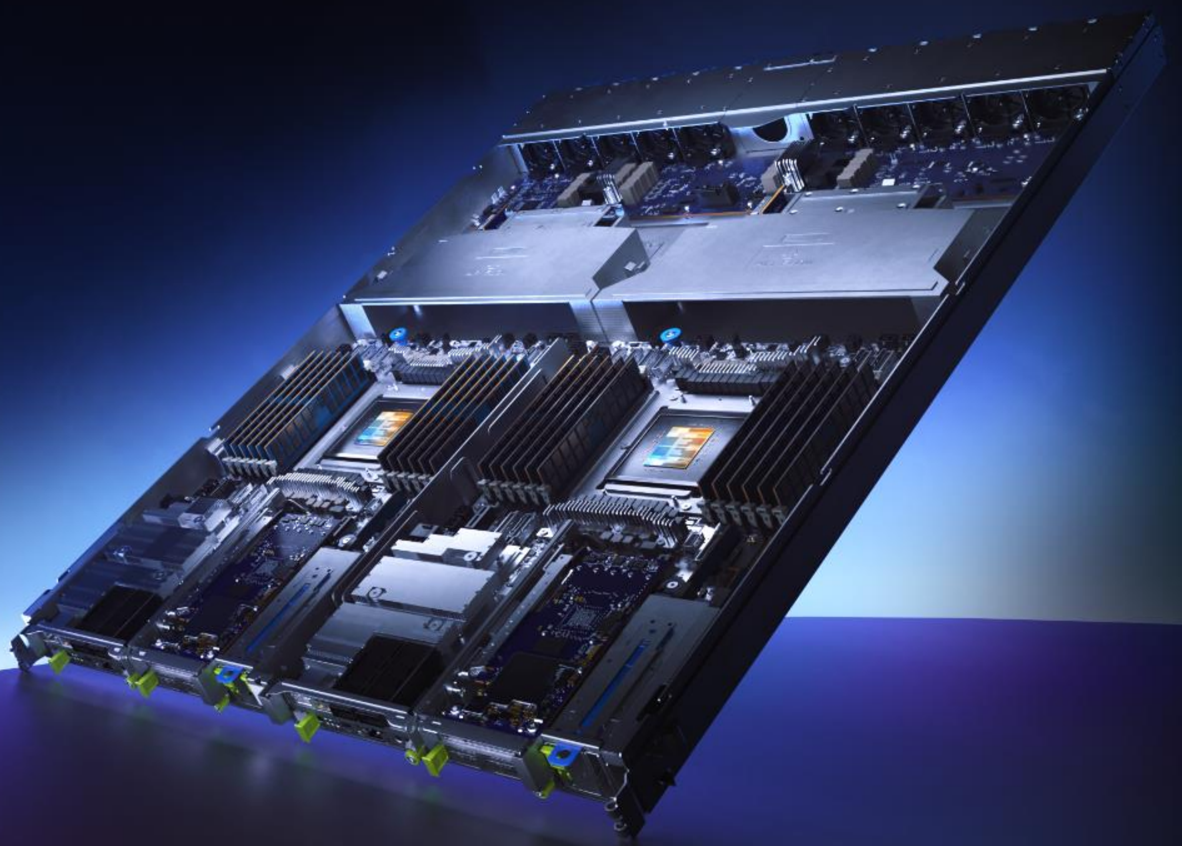
FRONT STORAGE (per node)
- 15mm Data Drive 7600 1.92TB
- 9.5mm Boot Drive 7600 1.92TB

INTERNAL STORAGE(per node)
- 9.5mm Data Drive 7600 1.92 TB
- E1.S Expansion Slot

MEMORY(per node)
- 12 DDR5 8800 DIMM Slots
- CXL 3.0

EXPANSION SLOTS (per node)
- PCIe x16 FHHL Gen 6.0 Slot
- PCIe x16 HHHL Gen 6.0 Slot
- Compute Express Link 3.0
- 200 Gb/s Ethernet Port

AIR COOLED
- 5 Dual Rotor Fans (per node)
- 5˚C to 35˚C Operating Temperature



Accelerated time-to-market with known-good reference design

Arm AGI CPU Reference servers

Arm AGI CPU 10U dual node
Reference server

Arm AGI CPU 2U2P
Reference server

Arm AGI CPU Commercial servers

Supermicro 5U
Arm AGI CPU
PCIe GPU system

Supermicro 2U Hyper
Arm AGI CPU Hyper
system

Supermicro 2U Hyper-E
Arm AGI CPU server

Supermicro 2U4N
Arm AGI CPU
Liquid-cooled server

Lenovo HR650a V3 2U Arm
AGI CPU system

ASRock Rack
2OU2N-Arm system

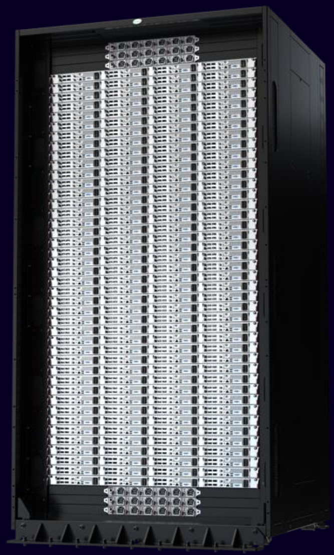
Arm AGI CPU Rack-scale systems

Reference 1U 2-Node
Arm AGI CPU
Rack-Scale solution

Air-cooled

Supermicro 2U 4-node
Arm AGI CPU
Rack-Scale solution

Liquid-cooled



Supermicro 1U 4-node
Arm AGI CPU
Rack-scale solution

Liquid-cooled

Extending cloud & AI software ecosystem to Arm AGI CPU

AI Agentic Workloads

Applications

Software Tools, Frameworks, Libraries

Hypervisors

Operating Systems

arm AGI CPU

Enablement steps

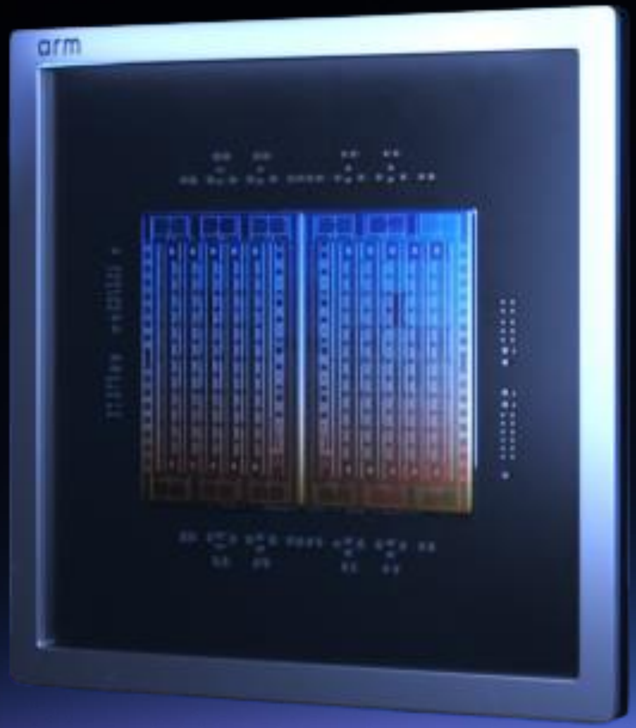
Provide platform access to partners

Validate specific software requirements

Workload benchmarking and optimizations

Enable customer and developer migrations

arm AGI CPU



Purpose-built for Agentic AI

- Balanced architecture focused on Agentic AI workloads
- Market-leading memory bandwidth per core and I/O bandwidth per core

Built to scale

- Rack is the new unit of compute
- Higher rack level compute density under given power envelope

Arm platform advantage

A unique continuum:
Architecture → IP → CSS → SoC

Proven technology

Built on Arm Neoverse, the technology used by leading cloud service providers and hyperscalers

Long-term commitment

The first product on a long-term Arm SoC roadmap for this market

Thank you!